Unit 9 End-of-Unit Assessment - Form A Teacher Key

8 items · 3 points each · 24 total. Award 1 for a valid model/setup, 1 for correct mathematics/units, and 1 for a requested justification. Use the applicable evidence when the task is conceptual; accept equivalent methods and response modes. Do not count writing length as mathematical mastery.

Item 1

Prompt: A rental costs $4 per hour. Identify independent and dependent quantities and explain.

Solution / evidence: Hours is input/independent; cost dependent; cost is computed from hours at 4 dollars/hour.

Item 2

Prompt: Use y=4x+5 for x=0,2,5. Explain the starting amount and input jumps.

Solution / evidence: 5,13,25; starting amount 5; change depends on the input jump.

Item 3

Prompt: For y=4x, give and plot the points at x=0,1,3; label axes.

Solution / evidence: (0,0),(1,4),(3,12); label context-independent input/output axes.

Item 4

Prompt: Notebooks cost $4 each. Interpret (2,8); is (2.5,10) a whole-notebook purchase?

Solution / evidence: Two notebooks cost $8. The fractional point fits the formula but is infeasible for whole notebooks.

Item 5

Prompt: For y=4x+5, compare x=2 and x=4. Find the change per added input unit; explain why y/x is not the rate.

Solution / evidence: Outputs 13 and 21; change 8 over 2 units gives rate 4. The fee is included in y/x.

Item 6

Prompt: Write an additive rule for (0,4),(2,6),(5,9), and check.

Solution / evidence: y=x+4; all rows reproduced.

Item 7

Prompt: A cost is $4 per item plus one $7 fee. Write its equation and evaluate 4 items.

Solution / evidence: c=4n+7; cost 23 dollars.

Item 8

Prompt: Items cost $4. Find minimum whole items to raise at least $41, and maximum affordable with that same budget.

Solution / evidence: Minimum 11 costs/raises 44; maximum 10 costs 40. Test both neighbors against 41.

Response: group errors by interpretation/model versus calculation versus explanation. Use the corresponding lesson Teaching Support Guide, revise one step, and reassess with its reserved fresh check. Do not reuse the already seen form as independent evidence.